

SRINITHIYA M

Thanjavur, Tamil Nadu, India | +91-9342700944 | srinithiyaofficial@gmail.com |

LinkedIn : <https://www.linkedin.com/in/srinithiya-m-081243291/> | GitHub : <https://github.com/srinithiyadev> |

Portfolio : <https://srinithiyadev.github.io>

PROFESSIONAL SUMMARY

DevOps and Site Reliability Engineer with hands-on experience designing and deploying cloud-native infrastructure on AWS. Proven ability to build scalable Kubernetes clusters on EKS, automate CI/CD pipelines with Jenkins and GitHub Actions, and implement real-time monitoring using Prometheus and Grafana. Proficient in Terraform, Ansible, Docker, and Python. Passionate about reducing operational overhead, improving deployment reliability, and building infrastructure that scales.

TECHNICAL SKILLS

Programming Languages : Python, Shell Scripting (Bash), SQL

Frontend : React

Cloud Platforms : AWS (EC2, S3, IAM, EKS, VPC), boto3

DevOps & CI/CD Tools : Docker, Docker Compose, Kubernetes (EKS, Minikube), Jenkins, Helm, GitHub Actions

Infrastructure as Code : Terraform, Ansible

Monitoring & Observability : Prometheus, Grafana, AWS CloudWatch

Databases : MySQL, PostgreSQL, Redis, MongoDB

Version Control & Registries : Git, GitHub, DockerHub

Operating Systems : Linux (Ubuntu, Fedora), Windows

Networking : TCP/IP, DNS, HTTP/HTTPS, Load Balancing, REST APIs

SOFT SKILLS

Communication | Team Collaboration | Problem Solving | Quick Learner | Adaptability | Troubleshooting | Attention to Detail | Time Management

PROJECTS

URL Shortener – Kubernetes DevOps Automation & Monitoring System

2026

Tech: Python (Flask), Docker, Kubernetes (AWS EKS), Jenkins, Prometheus, Grafana, DockerHub, PostgreSQL

- Deployed a scalable URL Shortener on AWS EKS using Kubernetes, handling auto-scaling and zero-downtime deployments.
- Implemented CI/CD pipeline with Jenkins, automating Docker image builds, DockerHub pushes, and EKS deployments on every code commit.
- Configured Prometheus scrape targets and Grafana dashboards for real-time application metrics, reducing incident detection time by 30%.
- Exposed the application via Kubernetes LoadBalancer service integrated with AWS networking for reliable external traffic routing.
- Containerized the Flask application with optimized Docker images, cutting deployment size and improving startup performance.

SRE Incident Management System

2026

Tech: FastAPI, MongoDB, PostgreSQL, Redis, React, Docker Compose, Prometheus, Grafana

- Built scalable IMS with Redis-based signal debouncing to reduce DB writes and improve performance under high traffic.
- Designed State Machine pattern for incident lifecycle (OPEN→INVESTIGATING→RESOLVED→CLOSED) with mandatory RCA enforcement before closure and auto MTTR calculation.
- Engineered async backend with rate limiting and Redis Streams for backpressure handling — improved stability under high load.
- Containerized 6 services with Docker Compose and integrated Prometheus + Grafana for real-time observability.
- Achieved 12/12 unit tests passing for RCA validation and state machine logic.

CloudOptima – Multi-Cloud Cost Monitoring Web Application

2025

Tech: Python, Flask, PostgreSQL, AWS (boto3), Brevo API, Render, Vercel

- Engineered a multi-cloud cost monitoring platform using AWS boto3, reducing manual cost tracking effort by 40%.
- Built a Flask REST API backend with PostgreSQL to store cloud resource data and expose endpoints for idle resource detection.
- Integrated Brevo API for automated contact management and bulk email alerting on cost threshold breaches.
- Deployed frontend on Vercel and backend on Render with CI/CD integration for automated release pipelines.
- Implemented structured logging and automated test suites to improve application reliability and accelerate debugging.

EDUCATION

Master of Science – Computer Science

2024 – 2026

Bishop Heber College, Tiruchirappalli, Tamil Nadu | Percentage: 76.50%

Bachelor of Science – Computer Science

2021 – 2024

PRIST University, Thanjavur, Tamil Nadu | Percentage: 76.10%

CERTIFICATIONS & PROFESSIONAL DEVELOPMENT

- Kubernetes: Your First Project – LinkedIn Learning (2026)
- Jenkins for Beginners – SimpliLearn (2025)
- Python Programming – Scaler (2025)
- Linux Fundamentals – Alison (2025)
- Docker for Developers – Le Wagon (2025)